

Day 18 Quiz– Q18 – Authentication-Authorization-JWT

1. What is the primary purpose of Authentication in an ASP.NET Core Web API?

- A. To determine what resources a user can access
- B. To verify the identity of a user
- C. To encrypt database records
- D. To validate HTTP request parameters

Answer: B.

2. What is the primary purpose of Authorization?

- A. Verifying the user's password
- B. Creating a JWT token
- C. Determining what an authenticated user is allowed to access
- D. Encrypting the HTTP request

Answer: C.

3. Which attribute is commonly used to protect an ASP.NET Core Web API controller or action?

- A. [Secure]
- B. [Authentication]
- C. [Authorize]
- D. [Protect]

Answer: C.

4. Which attribute allows an endpoint to be accessed without authentication when authorization is otherwise required?

- A. [AllowAnonymous]
- B. [NoAuthorize]
- C. [Public]
- D. [AnonymousAccess]

Answer: A.

5. Consider the following controller:

[Authorize]

[ApiController]

```
public class EmployeeController : ControllerBase
```

```
{
```

```
}
```

What is the default behavior?

- A. Only Admin users can access it
- B. Only Manager users can access it
- C. Authenticated users can access it, subject to additional authorization rules
- D. Anonymous users can access it

Answer: C

6. Which HTTP status code is normally returned when an unauthenticated user attempts to access an [Authorize] endpoint?

- A. 200
- B. 400
- C. 401
- D. 403

Answer: C.

7. Which HTTP status code generally indicates that the user is authenticated but does not have sufficient permission?

- A. 200
- B. 401
- C. 403
- D. 404

Answer: C.

8. Which HTTP header is normally used to send a JWT access token?

- A. Token
- B. Authentication
- C. Authorization
- D. JWT-Token

Answer: C.

9. Which format is commonly used when sending a JWT in the Authorization header?

- A. Authorization: JWT <token>
- B. Authorization: Bearer <token>
- C. Authorization: Token <token>
- D. Authorization: Basic <token>

Answer: B.

10. Which of the following represents the typical structure of a JWT?

- A. Header.Payload.Signature
- B. Header.Body.Password
- C. Username.Password.Signature
- D. Request.Response.Signature

Answer: A.

11. Why should a plaintext password generally NOT be stored in a database?

- A. It makes SQL queries slower
- B. It consumes too much disk space

- C. Anyone obtaining the database could directly see users' passwords
- D. ASP.NET Core does not support passwords

Answer: C.

12. Suppose the database contains:

Users

Id

Username

PasswordHash

Email

Role

During login, what should the application primarily do?

- A. Compare plaintext passwords stored in the database
- B. Compare the supplied password against the stored password hash using the appropriate password-verification mechanism
- C. Generate a JWT without validating the password
- D. Return the PasswordHash to the client

Answer: B

13. Is a JWT payload inherently encrypted?

- A. Yes, always
- B. No, JWT signing does not by itself encrypt the payload
- C. Yes, if HTTPS is used
- D. Yes, if the token contains claims

Answer: B

14. Which namespace commonly contains the JWT Bearer authentication implementation used by ASP.NET Core?

- A. Microsoft.AspNetCore.Authentication.JwtBearer
- B. Microsoft.AspNetCore.JWT.Core
- C. System.Web.JWT
- D. Microsoft.EntityFrameworkCore.JWT

Answer: A.

15. In a typical ASP.NET Core Web API pipeline, which order is appropriate?

- A. UseAuthorization() → UseAuthentication()
- B. UseAuthentication() → UseAuthorization()
- C. UseAuthorization() → UseRouting() → UseAuthentication()
- D. UseAuthentication() should never be used

Answer: B.

16. Which syntax allows either Admin or Manager to access an endpoint?

- A. [Authorize(Roles = "Admin,Manager")]
- B. [Authorize(Roles = "Admin|Manager")]
- C. [Authorize(Roles = "Admin && Manager")]
- D. [Authorize(Roles = "Admin + Manager")]

Answer: A.

17. What is a claim?

- A. A database table
- B. A piece of information about a user or identity
- C. A SQL stored procedure
- D. A password encryption algorithm

Answer: B.

18. Which property can be used to check whether the current request has an authenticated identity?

- A. User.Identity.IsAuthenticated
- B. Request.IsAuthenticatedUser
- C. User.IsLoggedIn()
- D. Identity.IsValid()

Answer: A.

19. Consider:

```
[Authorize]
[HttpGet]
public IActionResult GetProfile()
{
    ...
}
```

A valid JWT is supplied, but the token has expired. What should normally happen?

- A. The request is treated as authenticated
- B. The expired token should fail authentication
- C. The API automatically renews the token
- D. The database password is requested automatically

Answer: B.

20. Consider the following requirement:

"Only authenticated Admin users can delete employees. Managers can view employees but cannot delete them. Normal Users can only view their own profile."

Which combination provides the most appropriate design?

- A. Use only [AllowAnonymous] and check usernames manually
- B. Use JWT authentication with role/claim-based authorization and appropriate [Authorize] rules
- C. Store the Admin password in appsettings.json
- D. Allow all users to access all endpoints and hide buttons in the UI

Answer: B.